

SECTION 1 — CORE FOUNDATION

Table 1.1: **schools**

```
CREATE TABLE schools (  
  id BIGINT AUTO_INCREMENT PRIMARY KEY,  
  uuid CHAR(36) NOT NULL UNIQUE,  
  code VARCHAR(50) NOT NULL UNIQUE,  
  name VARCHAR(255) NOT NULL,  
  board VARCHAR(50),  
  address TEXT,  
  city VARCHAR(100),  
  state VARCHAR(100),  
  country VARCHAR(100) DEFAULT 'India',  
  pincode VARCHAR(10),  
  phone VARCHAR(20),  
  email VARCHAR(255),  
  website VARCHAR(255),  
  logo_url VARCHAR(500),  
  status ENUM('ACTIVE','INACTIVE') DEFAULT 'ACTIVE',  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  is_deleted BOOLEAN DEFAULT FALSE  
);
```

Table 1.2: **academic_years**

```
CREATE TABLE academic_years (  
  id BIGINT AUTO_INCREMENT PRIMARY KEY,
```

```
school_id BIGINT NOT NULL,  
code VARCHAR(20) NOT NULL,  
name VARCHAR(50) NOT NULL,  
start_date DATE NOT NULL,  
end_date DATE NOT NULL,  
is_current BOOLEAN DEFAULT FALSE,  
status ENUM('ACTIVE','CLOSED') DEFAULT 'ACTIVE',  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
is_deleted BOOLEAN DEFAULT FALSE,  
FOREIGN KEY (school_id) REFERENCES schools(id)  
);
```

SECTION 2 — ACADEMIC STRUCTURE

Table 2.1 : **classes**

```
CREATE TABLE classes (  
id BIGINT AUTO_INCREMENT PRIMARY KEY,  
school_id BIGINT NOT NULL,  
name VARCHAR(50) NOT NULL,  
category ENUM('PRIMARY','MIDDLE','SECONDARY','HIGHER'),  
sequence_order INT NOT NULL,  
is_final_class BOOLEAN DEFAULT FALSE,  
status BOOLEAN DEFAULT TRUE,  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
is_deleted BOOLEAN DEFAULT FALSE,  
FOREIGN KEY (school_id) REFERENCES schools(id)  
);
```

Table 2.2 : **sections**

```
CREATE TABLE sections (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    class_id BIGINT NOT NULL,  
    school_id BIGINT NOT NULL,  
    name VARCHAR(20) NOT NULL,  
    room_no VARCHAR(20),  
    capacity INT DEFAULT 60,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
    FOREIGN KEY (class_id) REFERENCES classes(id),  
    FOREIGN KEY (school_id) REFERENCES schools(id)  
);
```

Table 2.3 : **subjects**

```
CREATE TABLE subjects (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    school_id BIGINT NOT NULL,  
    name VARCHAR(100) NOT NULL,  
    code VARCHAR(20) NOT NULL UNIQUE,  
    subject_type ENUM('CORE','ELECTIVE','PRACTICAL','LAB') DEFAULT 'CORE',  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
```

```
is_deleted BOOLEAN DEFAULT FALSE,  
FOREIGN KEY (school_id) REFERENCES schools(id)  
);
```

Table 2.4 : **class_subjects**

```
CREATE TABLE class_subjects (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    class_id BIGINT NOT NULL,  
    subject_id BIGINT NOT NULL,  
    academic_year_id BIGINT NOT NULL,  
    is_optional BOOLEAN DEFAULT FALSE,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (class_id) REFERENCES classes(id),  
    FOREIGN KEY (subject_id) REFERENCES subjects(id),  
    FOREIGN KEY (academic_year_id) REFERENCES academic_years(id)  
);
```

SECTION 3 — USER & STAFF MANAGEMENT

Table 3.1 : **users**

```
CREATE TABLE users (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    uuid CHAR(36) NOT NULL UNIQUE,  
    school_id BIGINT NOT NULL,  
    email VARCHAR(255) NOT NULL UNIQUE,  
    phone VARCHAR(20),  
    password_hash VARCHAR(255) NOT NULL,  
    role ENUM('ADMIN','TEACHER','STAFF','STUDENT','PARENT') NOT NULL,
```

```
avatar_url VARCHAR(500),
last_login_at TIMESTAMP NULL,
is_active BOOLEAN DEFAULT TRUE,
is_deleted BOOLEAN DEFAULT FALSE,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
FOREIGN KEY (school_id) REFERENCES schools(id)
);
```

Table 3.2 : **staff**

```
CREATE TABLE staff (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,
    user_id BIGINT UNIQUE,
    school_id BIGINT NOT NULL,
    employee_code VARCHAR(50) NOT NULL UNIQUE,
    name VARCHAR(150) NOT NULL,
    designation VARCHAR(100),
    qualification VARCHAR(255),
    joining_date DATE,
    salary DECIMAL(10,2),
    experience_years DECIMAL(4,1) DEFAULT 0,
    status ENUM('ACTIVE','RESIGNED','TERMINATED') DEFAULT 'ACTIVE',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id),
    FOREIGN KEY (school_id) REFERENCES schools(id)
);
```

SECTION 4 — STUDENT LIFECYCLE

Table 4.1 : **students**

```
CREATE TABLE students (  
  id BIGINT AUTO_INCREMENT PRIMARY KEY,  
  uuid CHAR(36) NOT NULL UNIQUE,  
  school_id BIGINT NOT NULL,  
  admission_no VARCHAR(50) NOT NULL UNIQUE,  
  first_name VARCHAR(100) NOT NULL,  
  last_name VARCHAR(100),  
  gender ENUM('MALE','FEMALE','OTHER') NOT NULL,  
  date_of_birth DATE NOT NULL,  
  blood_group VARCHAR(5),  
  phone VARCHAR(20),  
  email VARCHAR(255),  
  photo_url VARCHAR(500),  
  admission_date DATE,  
  category ENUM('GENERAL','OBC','SC','ST','EWS'),  
  status ENUM('ACTIVE','PASSED_OUT','DROPPED','TRANSFERRED') DEFAULT 'ACTIVE',  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  is_deleted BOOLEAN DEFAULT FALSE,  
  FOREIGN KEY (school_id) REFERENCES schools(id)  
);
```

Table 4.2 : **student_enrollments**

```
CREATE TABLE student_enrollments (  
  id BIGINT AUTO_INCREMENT PRIMARY KEY,
```

```

student_id BIGINT NOT NULL,
academic_year_id BIGINT NOT NULL,
class_id BIGINT NOT NULL,
section_id BIGINT NOT NULL,
roll_no VARCHAR(20),
enrollment_date DATE NOT NULL,
status ENUM('ENROLLED','PROMOTED','DETAINED','PASSED_OUT') DEFAULT 'ENROLLED',
is_current BOOLEAN DEFAULT TRUE,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
FOREIGN KEY (student_id) REFERENCES students(id),
FOREIGN KEY (academic_year_id) REFERENCES academic_years(id),
FOREIGN KEY (class_id) REFERENCES classes(id),
FOREIGN KEY (section_id) REFERENCES sections(id)
);

```

Table 4.3 : guardians

```

CREATE TABLE guardians (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,

    user_id BIGINT,
    name VARCHAR(150) NOT NULL,

    relation ENUM(
        'FATHER',

```

```
'MOTHER',
'GUARDIAN',
'OTHER'
) NOT NULL,

phone VARCHAR(20) NOT NULL,
email VARCHAR(255),

occupation VARCHAR(100),
annual_income DECIMAL(12,2),

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT fk_guardian_user
FOREIGN KEY (user_id) REFERENCES users(id)
);
```

Table 4.4 : **student_guardians**

Many-to-many

Supports primary + emergency contacts

```
CREATE TABLE student_guardians (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,

    student_id BIGINT NOT NULL,
    guardian_id BIGINT NOT NULL,
```


is_primary BOOLEAN DEFAULT FALSE,

is_emergency BOOLEAN DEFAULT FALSE,

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT fk_sg_student

FOREIGN KEY (student_id) REFERENCES students(id),

CONSTRAINT fk_sg_guardian

FOREIGN KEY (guardian_id) REFERENCES guardians(id),

UNIQUE KEY uq_student_guardian (student_id, guardian_id)

);

Table 4.5 : **student_documents**

CREATE TABLE student_documents (

id BIGINT AUTO_INCREMENT PRIMARY KEY,

student_id BIGINT NOT NULL,

document_type ENUM(

'AADHAR',

'BIRTH_CERT',

'MARKSHEET',

'PHOTO',

'OTHER'

```
) NOT NULL,  
  
file_url VARCHAR(500) NOT NULL,  
  
is_verified BOOLEAN DEFAULT FALSE,  
verified_by BIGINT,  
  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  
CONSTRAINT fk_doc_student  
    FOREIGN KEY (student_id) REFERENCES students(id),  
  
CONSTRAINT fk_doc_verified_by  
    FOREIGN KEY (verified_by) REFERENCES users(id)  
);
```

SECTION 5 — ATTENDANCE

Table 5.1 : **student_attendance**

```
CREATE TABLE student_attendance (  
    attendance_id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    student_id BIGINT,  
    section_id BIGINT,  
    attendance_date DATE NOT NULL,  
    status ENUM('PRESENT','ABSENT','LEAVE','LATE') NOT NULL,  
    remarks VARCHAR(255),
```

```
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
FOREIGN KEY (student_id) REFERENCES students(id),  
FOREIGN KEY (section_id) REFERENCES sections(id)  
);
```

Table 5.2 **staff_attendance**

```
CREATE TABLE staff_attendance (  
attendance_id BIGINT AUTO_INCREMENT PRIMARY KEY,  
staff_id BIGINT,  
attendance_date DATE NOT NULL,  
status ENUM('PRESENT','ABSENT','LEAVE','LATE') NOT NULL,  
remarks VARCHAR(255),  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
FOREIGN KEY (staff_id) REFERENCES staff(id)  
);
```

SECTION 6 — EXAMS & RESULTS

Table 6.1 : **exams**

```
CREATE TABLE exams (  
id BIGINT AUTO_INCREMENT PRIMARY KEY,  
name VARCHAR(150) NOT NULL,  
academic_year_id BIGINT,  
class_id BIGINT,  
section_id BIGINT,  
exam_start_date DATE NOT NULL,  
exam_end_date DATE NOT NULL,
```

```
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
FOREIGN KEY (academic_year_id) REFERENCES academic_years(id),  
FOREIGN KEY (class_id) REFERENCES classes(id),  
FOREIGN KEY (section_id) REFERENCES sections(id)  
);
```

Table 6.2 : **exam_subjects**

```
CREATE TABLE exam_subjects (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    exam_id BIGINT,  
    subject_id BIGINT,  
    total_marks INT NOT NULL,  
    passing_marks INT NOT NULL,  
    FOREIGN KEY (exam_id) REFERENCES exams(id),  
    FOREIGN KEY (subject_id) REFERENCES subjects(id)  
);
```

Table 6.3 : **exam_results**

```
CREATE TABLE exam_results (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    exam_id BIGINT,  
    student_id BIGINT,  
    subject_id BIGINT,  
    marks_obtained DECIMAL(5,2) NOT NULL,  
    grade VARCHAR(5),  
    remarks VARCHAR(255),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (exam_id) REFERENCES exams(id),
```

```
FOREIGN KEY (student_id) REFERENCES students(id),  
FOREIGN KEY (subject_id) REFERENCES subjects(id)  
);
```